The Socio-Political Impact of Game Design: How does Cities: Skylines Manage its Identity as a Video Game and an Urban Planning Simulation

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Abstract

This paper examines the commercial virtual simulation video game, “Cities: Skylines” as a video game and as a simulation separately, and their relationship with each other and the subsequent socio-political impact on its players. The analysis consists of urban planning framework developed in the 20th century, as well as literature regarding Ludology. The focus of the analysis is on the systems and gameplay mechanics, and its influence on its players. While the term simulation would suggest ‘realistic’ systems that mimic that of real life, this analysis focuses on the games’ identity as a video game, and the subsequent impact on its accuracies as a city simulation.

Introduction

Cities: Skylines is a city-building game released in 2015, developed by Colossal Order and published by Paradox Interactive. It is a single player open-ended city-building simulation, and a spiritual successor to Maxis and EA’s Sim City series. The game is an urban planning simulator, with players engaging with different aspects such as zoning, roads, taxation, public services and transportation, industries, etc. They must manage a variety of resources and elements, including budget, health, employment, traffic, pollution and natural resources. The game also provides a sandbox mode, allowing players to treat it as a creative tool. The game also has massive support and documentation for modding, allowing players to create their own assets as well as system modifications to the game, varying from quality-of-life adjustments to complete overhauls of the game’s systems.

The game works on 3 principal zoning demands; Residential, Commercial, and Industrial (RCI). The player must maintain demand for all 3 zoning types to continue progressing in the game. It is possible for players to ‘soft lock’ (an inadvertent game over where the player, due to either a fault in the game’s systems or the player’s own actions, can lead to an earlier unintended ending for the playthrough) themselves in earlier stages due to mismanaging their RCI’s. This aspect requires a fundamental understanding of supply and demand mechanics, as the game is primarily designed around its economical and traffic systems. Most media, when consumed, has some form of social or political ramifications, no matter how apolitical it might seem. City simulators have been (in)famous in their use in real world politics, such as the 1990 Democratic primary elections in Providence, Rhode Island (Koebler), to being used in educational institutes such as schools.

Early Game Interactions

The paper henceforth will postulate the player playing the game on its default settings, on the latest version of the game, with all ‘vanilla’ DLCs and expansions. The game starts off with a singular plot of land, equivalent to a 2x2km area, along with an interchange exit from a nearby highway, which acts as the hypothetical connection to the rest of the world, and the input/output for the first period of the game. This forms the basis for the player’s interaction with the game’s systems. The game works within the confines of industrial production, and henceforth the city becomes a system of capital flow, individuals live (Residential) to produce (Industrial) and to consume (Commercial).

The player starts off slow, with modicum increments in housing, commercial, and industrial zoning, and is perhaps the tensest period of the game. Due to the almost non-existent income, and lack of land area and resources at the beginning, players must be careful to manage the limited budget of 70,000 ₵ (cells) the game starts them off with. This development shapes the starting establishment and can create the entire fundamental of the player's city as they move forwards.

The player is also limited to their access to tools and 'blueprints' available in the first 'act' of the game, so they work with relatively cheap and inefficient buildings and infrastructure, which can be replaced with larger, more efficient versions of the same, once the player has progressed into the game and unlocked some of the later buildings. The game's progression is directly linked to the city's population, having levels such as Little Hamlet, Tiny Town, to Small City, and the final being Megalopolis, a milestone cap at 90,000 citizens, but the game can go on indefinitely. Players unlock more tools, items and blueprints as they increase the size of the city.

Zoning

Following on the footsteps of its predecessors, *Cities: Skylines* also adopts the RCI demand mechanic from previous city simulations, notably Sim City. As mentioned previously, the city functions on these 3 principal zoning demands. This provides the basis for the gameplay loop for every city. The player goes through cycles of demand in their playthrough, moving between moments of extremely high demands to none at all. This is represented visually in the bottom tab that houses the game's functions, as 3 bars; green representing residential, blue being commercial and orange representing industrial demand. This demand is controlled by a variety of factors, such as job availability, land value, competition, connectivity, municipal services, etc.

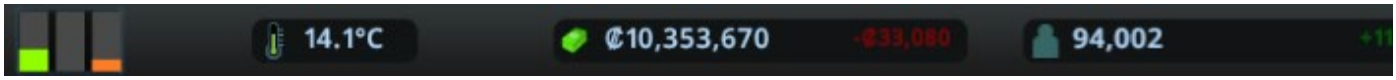

Figure 1 Screenshot from Cities: Skylines of part of the game's UI, the 3 demand bars are present on the left.

While some factors are available to the player, their effect on the demands is intentionally obscure, encouraging players to analyse their city and experiment with their decision making on what could be the right call for the city.

How the Zoning System Works

Cities provides the player with two main forms of development. This paper will categorize them as private and state development. One of the game's fundamental rules requires all buildings to be constructed by a roadside.

When the player builds a road, if there is space available, the game will generate zoning grids on its sides, following the entire length of the road, and going out a maximum of 4/5 (confirm this) grid spaces. Any building placed, whether by the player or the game itself, will align itself according to this grid (although the player can bypass this restriction using mods.

If a prospective zoning space is obstructed, by environmental factors, such as rocks or ground elevation, or other roads, the game will omit the squares affected. Grids also generate parallel to the road placed, so if two roads intersect in acute angles, the game gives zoning preference to the road placed last. When roads intersect in right angles, and are appropriately spaced, 8zu, (zone units) 4 on each side, the game forms a city 'block' and fills in the space entirely.

Figure 2 Zoning depicted in Cities: Skylines, coloured grids represent their respective zoning type

This mechanic indirectly encourages grid planning, as when paired with the limited space available, players are naturally inclined to be as efficient with their space as possible, and the most convenient road structures and grid-based planning tends to be the most efficient solution for first time players. Since every building aligns itself perpendicularly to its road, this creates rigid city patterns that don't account for aesthetic or functional attributes. This constricts player creativity as one must go out of their way to break up this structure, but they remain unrewarded by the game's systems for this effort, as these changes only affect the painted reality of the game, and there is no incentive other than player aesthetics to implement these developments. In fact, the player may end up losing critical zoning space to accommodate their aesthetic choices, and at worst may even be detrimental to the function of their city.

Private and State Development

As mentioned earlier, any developmental project in the player's city can be divided into either privately funded development or state projects. Private development in the game refers to any development not directly established by the player. This includes residential buildings, offices, small factories, shops, etc. The zoning grid is essentially a placeholder for every possible development site. The player can either designate a block or an area on a zoning grid to a particular RCI, after which, if the respective demand exists, the game will automatically fill in those spaces with buildings. I.e. if the player designates an area as residential zoning, cims will move in and start building houses in the available space until every spot is filled. The player can set an architectural style for that district, policies varying from traffic laws to building regulations, but beyond that do not have much creative control over them. Private development

is autonomous, and the player has minimal control over them, but is more efficient for the flow of the game, as it allows players to develop large areas of land without placing every piece manually.

State or municipal development on the other hand, refers to services or specialized buildings the player manually places on available lots. These buildings also abide by the zoning rules, but the player can select their specific position within the grid. Once again, these buildings must also have road access. These can range from municipal services such as police and fire stations, waste management, recreational areas such as parks and plazas, larger state-owned industries. Every building is placed manually by the player, and its position is crucial to the functioning of the city. Municipal buildings have a range they cover, and this is determined by a defined radius around it, as well as connectivity through road, although services will still serve citizens outside this range, at the cost of manpower and the time it takes to reach them. This encourages players to section their cities into districts, as it helps manage services once they expand.

Residential Zoning

Housing in *Cities: Skylines* is determined entirely by the zoned grids created by the player. While the player can create municipally owned commercial and industrial buildings, housing is the only demand that cannot be placed manually by the player and depends entirely on the game's system.

Housing is classified into two subcategories: low- and high-density housing. Low density housing is occupied by families, i.e. adults and kids. Hence, placing low density housing, encourages long term city population growth, at the expense of using up more land. As families move into these homes, they're encouraged to have kids and in turn grow the population size. When starting the game, the player will only have access to low density residential zoning, so inner cities generally tend to be made up of smaller suburban housing plots, and the player later expands into higher density zoning on the peripheries as they progress and start to run out of land. This can lead to a form of inner-city decay, as the player is incentivized to develop newer land, as they will have unlocked better buildings as well, and ignore the low-density areas in favour of development in high density neighbourhoods, to maximize the efficiency of newer developments to impact as many of their cims as possible. Because all data is presented as an aggregate, it encourages players to maximize efficiency and overlook underrepresented neighbourhoods.

High density housing, on the other hand, is favoured by young adults and college students, and offer greater immediate population influx, at the cost of internal population growth, as people living in apartments are not likely to have kids. However, for a player, this change in priority between housing types are often overlooked, as all population input is presented to the player as a "+/-*p*" next to their total population bar, where *p* represents the number of people moving in each week. Although the player can also access their city's average birth/death rate, the game does not present this data to the player actively and does not help them to internalize the effects of these changes either. Hence, when the player zones high density housing, they see *p* go up, and attribute that positive number to high density housing. While there are considerable changes to the gameplay by making the choice between type of housing, the weight of the decision is not really felt by the average player, and the game's UI design is inadvertently biased towards high density housing and limits the player's autonomy in their choice.

Land Value and Class

Class is minimally represented in the game. While there is a rough outline for income levels, there is no system to account for every cim/group and their individual needs, however, that can be justified as limits in the technicalities of the medium. Class is represented in the game as building 'levels', residential having levels up to 5, and apply to all types of housing.

Residential levels are determined by two factors, land value and the average education level of the citizens in that area. Amenities such as parks and plazas, access to municipal services, shorelines, all provide increased land value. While the two may seem apparently similar, happiness and land value are two separate systems, and a cim can be happy even in a low land value neighbourhood. Land value can be negatively affected by factors such as pollution or crime rate, so a neighbourhood near a landfill or highway will have a lower land value than one next to a beach.

Figure 3 Land Value visualized

Since the system is exponential, there is no detrimental effect to excessive land values, or excessive infrastructure as well. The game encourages the player to make have high land value, as that 'levels up' buildings. There is no concept of affordability, as, if demand exists, there will always be a new cim to fulfil that space. Cims can move out if they are unhappy, but it is by their own means. Cims adapt to rising costs and rent, as it's implied that the level of the building represents the cims residing within.

Similarly, there are no downsides to excessive implementation of services and amenities either. For example, police stations are a municipal service players can place to alleviate the crime rate in a neighbourhood, and in turn, provides a boost to the land value of its locality. A player can place as many stations as their budget allows, with no social repercussions. Police stations respond to crime 'requests', represented as an icon, indicating a rise in crime in the building it originates from. The station sends a car that picks up the criminal cim from the building, after which the notification is resolved. If the problem is not resolved, it can lead to the building

being abandoned entirely. The chance of crime occurring is determined by the general city unhappiness, and unemployment rate. This, however, is not communicated to the player, and once again, the inadvertent result is that the player reacts to the crime notification, and they build more police stations to respond to them, turning their city into a police state. On a first glance, this could be considered a successful implementation of a 'simulation', but it still lacks the real-world repercussions of having overfunded police.

Industrial Zoning and Financialization

Industrial demand is satisfied by two different zoning types, industrial and office buildings. Cities, in the eyes of the simplified demand format of city simulators, become spaces of capital flows. All these developments exist to serve one thing, production. Residents go to work at industrial zones, to produce goods to be sold by consumer outlets, which create the capital required to continue this system. The game, however, does not allow the player much space to escape this cycle, as all development is inherently linked to production. Henri Lefebvre identifies a triad of 'space production', in his book, 'The Production of Space' (Lefebvre). There are three levels of each social space, the physical space itself, its representation and the actual human experience that forms the lived space. The physical space is determined by actors; developers, state and architects. The secondary, representation of space, is the discourse that surrounds a space. This could be a local newsletter, a homeowner association, or any symbolic value given to the space by its inhabitants. The lived space comprises of the real interactions that occur between the people and the space itself, and without the lived experience the space cannot be considered a social space.

However, in *Cities: Skylines,* there is no scope for either of the latter two. Space is determined by capitalist production, and do not account for the lived experience as its not measured in any aspect of the game. In its identity as a video game, of a play space, it abandons the 'simulation' by Caillois' definitions (Caillois), to appeal to the player's desires of 'chance' and 'competition' instead. The player is instead trying to compete in the virtual market of the game's hidden system, and produces space that would benefit capitalist production, to fund the production of more space.

Industrial zones in the game work to provide jobs and produce goods to be sold at commercial locations. Hence, the player's city is reduced into two forms, residential and industrial, and their remainder is shaped by the relationship between the two, i.e. commercial. Even though the game provides parks and pedestrian districts to appeal to the lived experience, these are fictional representations of the real lived space, even within the reality of the game, as they only truly serve the land value of the space. For example, while pedestrian only districts are a feature implemented in the game, their actual benefits are reduced to an increase in land value, as effects from traffic such as air pollution are not measured by the game's system. In fact, the game makes many such choices to mitigate the negative causality caused by motor vehicles to accommodate their use. Pedestrian zones are also much more expensive to maintain, as a balancing mechanic, to discourage players from turning their entire city into a pedestrian hub. Pedestrian districts do not reflect the social and lived benefits of having walkable cities.

Figure 4 An example of a pedestrian path with a trolley line running through

Despite the emphasis on production, the player can omit industrial zones completely once in later stages and develop office zones instead. Unlike industrial zones, which create factories which in turn create ground pollution (the game having no metric for measuring air pollution), offices are far more sustainable and provide a better 'quality' of jobs. Offices exist solely to provide jobs (Paradox Interactive), and have no impact into the production pipeline itself, other than shifting higher educated citizens from factories into comfier skyscrapers. Any outlying demand for goods is filled automatically by the hidden import/export mechanic, which allocate money on its own from the player's budget to continue providing existing businesses with products to sell. Therefore, even if the player actively tries to disrupt the capital flow, the game will autocorrect itself on its own.

Traffic

The game's systems are based on its abilities as a traffic simulation. One of the key features highlighted by the publishers during its launch is its 'live' (Paradox Interactive) traffic system. Rather than creating models of cars used to represent the 'idea' of traffic, each car in the game has a physical presence in the traffic system. Each of them has a starting point and destination. The game's system is entirely dependent on motor for processing every flow in the game. As discussed earlier, buildings are based around roads, and vehicle access is crucial to their function, as every system and service is reliant on motor vehicles.

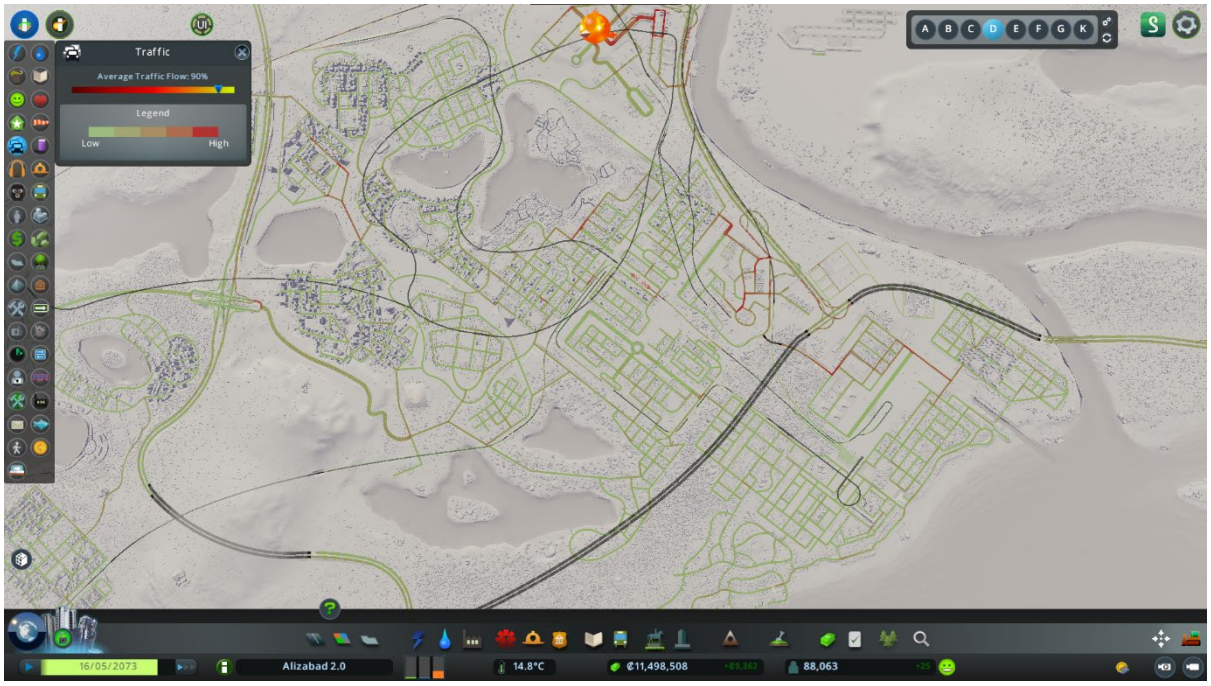

Figure 5 Looking at the Traffic overlay of a city

However, this is not the case in real life. Businesses in *Cities: Skylines* are supplied entirely through trucks. While there are alternative cargo options, such as cargo trains and boats, all good are eventually carried through trucks, whether they are larger big box stores or even smaller family-owned businesses. Cargo train stations are quite limited as well. They can only serve one train at any given time and release a series of trucks that go to different factories/businesses, even if they deliver to the same neighbourhood. Similarly, when industrial buildings, including larger state-owned development, are all treated as individual buildings, despite being part of the same larger 'company', and have individual trucks serving them.

Figure 6 Traffic in a State-Owned Agriculture Industry

In figure 6, despite all farmlands being part of the same 'farm', every farmland has its own set of delivery vehicles, rather than having the option to create a distribution centre to set up delivery sites that only use a fixed number of trucks to serve that entire section of the farm. For example, companies such as Tesco have distribution centres with rail lines running through them which then deliver to local distributors and stores (Tesco).

As mentioned earlier, in order to accommodate cars, the game's developers have made choices that directly impact the negative externalities that would come with an overreliance on motor vehicles. In an interview with Stone Librande, Sim City 2013's lead designer, Librande says "[parking] was kind of a problem, because we were originally just going to model real cities, but we quickly realized there were way too many parking lots in the real world and that our game was going to be really boring if it was proportional in terms of parking lots". While not the same game, it's safe to assume that the developers at Colossal Order were following a similar line of thought, as despite their claims to create a realistic traffic simulation, the game technically treats cars and cims as the same entity. An example of this is visible when a cim leaves their car; once they get out, the car seems to disappear behind them with no trace, as if they were in the driver's pocket. While the player can see cars parked alongside their roads, this is more of an aesthetic representation, as a vast majority of parked cars just disappear once they are not in use. Parking is not even a consideration for the player, despite vehicles playing such a large part in their cities. In the U.S, parking accounts for one-fifth of all land in city centers (Jacobs). This kind of omission fails to show the player the consequences of excessive motor vehicle dependency and encourages building more road infrastructure to combat the excessive traffic that are a 'requirement' for the functioning of the city. This however, leads to the phenomenon of induced demand, where by increasing the capacity of a road network, it leads to an eventual increase in traffic volume, as when driving becomes easier, more people are encouraged to use their cars, and the cycle continues (Hymel, Small and Dender).

Cims

The player is encouraged to see their citizens not as people, but as resources. There is no larger governing body, no local movements, no impactful representations of race and culture. The only indicator to the citizens welfare is a round 'face' icon near the population count, turning red and unhappy to indicate dissatisfaction, and a green smiling face to indicate contentment. Beyond this, the only communication between the player and their citizens is through the game's in-world Twitter/X, 'Chirper'. However, it mostly acts as a pretext to remind the player about issues in the cities or congratulate them on progress made in the game.

Disciplinary Societies

Michel Foucault introduces the term of *disciplinary societies*, in his 1975 book, Discipline and Punish, that addresses changes in society in the industrial age (Foucault). He claimed that "discipline proceeds from the distribution of individuals in space", and they are maintained by 'enclosures' the individual passes through, whether the school, the barracks or the factory. Under a disciplinary society, people are moved between these enclosures, their lives under constant observation when under the authority of these institutions. Therefore, the individual becomes a docile body, weary of observation, and limits the attempts to supersede the current authority, not out of fear of capital punishment, but just the unverifiable threat of observation.

The player is all observing *Panopticon* of their city. Due to the nature of the game, the player has unrestricted access to all aspects of their city and the lives of their citizens and is even physically separated from the city in the bird's eye perspective they have. This physical separation disconnects the player from the lived experience from the city, as they see the larger monumental architecture, the *ouvre* of the city instead of the view from the street. The player can also track a cim's movements, all aspects of their life, their family, their job, etc.

Similarly, the cims in a player's city, are these docile bodies. While not reacting out of fear, they are essentially products of *disciplinary society,* a perfect capitalist citizen, that pays their taxes, abides by the law, lives in accordance with the habits of flows, production and consumption supply and demand, from industrial capital to commodities of leisure. The player can choose to demolish a cim's house or place of business, and in such a case, they will simply relocate.

Cims do not have the ability to protest, to implement change, or have any say in how their city and living spaces are constructed. In fact, the cim cannot even interact with their city, as unlike the player's painted reality, with architecture, trees and lakes, the cim only interacts with their world instrumentally, seeing the city as a collection of paths and nodes that represent various variables. As Gilles Deleuze describes, in *societies of control*, an evolution of disciplinary societies "Individuals have become '*dividuals*,' and masses, samples, data, markets, or '*banks*'" (Deleuze). The cim is the dividual. Rather than a whole, complete, individual, the cim is instead a series of data sets that serve a variety of institutions.

Conclusion

City simulators hold an immense responsibility in shaping the development of its player's thinking on urban planning, especially after their increased use in education in city planning and development. As such, these simulations should reconsider their identity as games or be

able to affectively separate and communicate the differences between the two, in order to be effective as an educational tool.

There are solutions to implementing lived experiences within the game system. Rather than a focus on capitalist production, developers could reconsider the goals of the player to accommodate the happiness of their citizens in a more meaningful way. Having the option to explore the city, interact with citizen, set up local governments and representatives, will allow the player to form a relationship with their citizens. Restructuring demand to highlight citizen needs would encourage a natural path towards a conducive city.

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